WORKSHEET 4.6 : INTERSECTION OF 3 PLANES

8a) $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 12 \\ 2 & -1 & 1 & 5 \\ 3 & 1 & -2 & 1 \end{array}\right] \xrightarrow[3R1-R3]{2R1-R2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 12 \\ 0 & 5 & 1 & 19 \\ 0 & 5 & 5 & 35 \end{array}\right] \xrightarrow{R3-R2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 12 \\ 0 & 5 & 1 & 19 \\ 0 & 0 & 4 & 16 \end{array}\right]$

$\xrightarrow[\frac{1}{4}R3]{5R1-2R2} \left[\begin{array}{ccc|c} 5 & 0 & 3 & 22 \\ 0 & 5 & 1 & 19 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow[R2-R3]{R1-3R3} \left[\begin{array}{ccc|c} 5 & 0 & 0 & 10 \\ 0 & 5 & 0 & 15 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow[\frac{1}{5}R2]{\frac{1}{5}R1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$

∴ P.O.I. $(2, 3, 4)$

b) $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 2 & -2 & 4 & 7 \\ 3 & -3 & 6 & 11 \end{array}\right] \xrightarrow[3R1-R3]{2R1-R2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$ } no solution!

3 planes are parallel & distinct

Since $\vec{n_1} = (1, -1, 2)$
$\vec{n_2} = (2, -2, 4)$
$\vec{n_3} = (3, -3, 6)$

but cartesian equations not scalar multiples

c) $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 5 \\ 2 & 2 & -4 & 6 \\ 1 & 1 & -2 & 3 \end{array}\right] \xrightarrow[R2-2R3]{2R1-R2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 5 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2}R2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$

↳ infinite soln

$\xrightarrow{R1+R2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 7 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$ ∴ $z = 2$, Let $y = t$, $x + y = 7$, $x = 7 - t$

∴ Line of intersection.

$(x, y, z) = (7, 0, 2) + t(-1, 1, 0), t \in \mathbb{R}$

2 coincident plane ≠ 1 intersecting plane

d) $\left[\begin{array}{ccc|c} -2 & 4 & 6 & -2 \\ 4 & -8 & -12 & 4 \\ 1 & -2 & -3 & 1 \end{array}\right] \xrightarrow[\substack{2R1+R2 \\ R1+2R3}]{-\frac{1}{2}R1} \left[\begin{array}{ccc|c} 1 & -2 & -3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ all the same planes.

$\therefore$ 3 concident planes, plane of intersection:

$$x-2y-3z-1=0$$

e) $\left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 1 & -1 & -2 & 5 \\ 3 & 3 & 6 & 5 \end{array}\right]$

$\vec{n_1} \parallel \vec{n_2}$ but eqn not scalar multiples,

$\therefore$ no solution.

2 parallel & distinct planes, 1 intersecting

f) $\left[\begin{array}{ccc|c} 1 & 3 & 5 & 10 \\ 2 & 6 & 10 & 18 \\ 1 & 3 & 5 & 9 \end{array}\right]$

$\vec{n_1} \parallel \vec{n_2} \parallel \vec{n_3}$

but only 2 equations scalar multiples

$\therefore$ 2 concident & 1 parallel & distinct planes

$\therefore$ no solution.

g) $\left[\begin{array}{ccc|c} 1 & -3 & -2 & 9 \\ 1 & 11 & 5 & -5 \\ 2 & 8 & 3 & 4 \end{array}\right] \xrightarrow[2R1-R3]{R1-R2} \left[\begin{array}{ccc|c} 1 & -3 & -2 & 9 \\ 0 & -14 & -7 & 14 \\ 0 & -14 & -7 & 14 \end{array}\right] \xrightarrow[R2-R3]{-\frac{1}{7}R2} \underbrace{\left[\begin{array}{ccc|c} 1 & -3 & -2 & 9 \\ 0 & 2 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]}$ infinite solution as a line!

$\rightarrow \overset{2R1+3R2}{\left[\begin{array}{ccc|c} 2 & 0 & -1 & 12 \\ 0 & 2 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]}$

$\therefore$ Let $z=t$

$2y+z=-2$

$2y=-2-t$

$y=-1-\frac{1}{2}t$

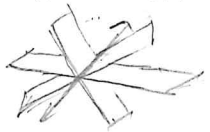
$2x-z=12$

$2x=12+t$

$x=6+\frac{1}{2}t$

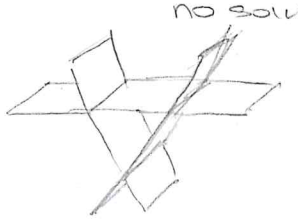
$\therefore (x,y,z)=(6,-1,0)+t\left(\frac{1}{2},-\frac{1}{2},1\right) \quad t\in\mathbb{R}$

Since $\vec{n_1} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$, $\therefore$ STAR-SHAPED. (NO-PLANAR NORMALS)



h)
$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 1 & -1 & -4 & -2 \\ 3 & 5 & 12 & 27 \end{array}\right] \xrightarrow[3R1-R3]{R1-R2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 0 & 2 & 6 & 8 \\ 0 & -2 & -6 & -9 \end{array}\right] \xrightarrow{R2+R3} \underbrace{\left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 0 & 2 & 6 & 8 \\ 0 & 0 & 0 & -1 \end{array}\right]}_{\text{no solution.}}$$

Since $\vec{n_1} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$, ∴ planes form a triangular prism. & no solution.



i)
$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & -2 & -3 & 0 \\ 3 & 2 & 4 & 0 \end{array}\right]$$

↑ Since all constant $D = 0$.
all planes go through the origin !!
and since $\vec{n_1} \not\parallel \vec{n_2} \not\parallel \vec{n_3}$, P.O.I. $(0,0,0)$

check.
$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 5 & 7 & 0 \\ 0 & -8 & -13 & 0 \end{array}\right] \xrightarrow[8R2+5R3]{5R1-R2} \left[\begin{array}{ccc|c} 10 & 0 & -2 & 0 \\ 0 & 5 & 7 & 0 \\ 0 & 0 & -9 & 0 \end{array}\right]$$

$$\xrightarrow[-\frac{1}{9}R3]{\frac{1}{2}R1} \left[\begin{array}{ccc|c} 5 & 0 & -1 & 0 \\ 0 & 5 & 7 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow[R2-7R3]{R1+R3} \left[\begin{array}{ccc|c} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow[\frac{1}{5}R2]{\frac{1}{5}R1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

∴ P.O.I. $(0,0,0)$

a)
$$x - 2y - z = 0$$
$$x + 9y - 5z = 0$$
$$k - y + z = 0$$

$\vec{n_1} = (1, -2, -1)$
$\vec{n_2} = (1, 9, -5)$
$\vec{n_3} = (k, -1, 1)$

since there's no $k \in \mathbb{R}$ s.t. any of these would be scalar multiples.
∴ to have a line of intersection, the normals must be coplanar

$$\vec{n_3} \cdot (\vec{n_1} \times \vec{n_2}) = 0$$
$$(k, -1, 1) \cdot (19, 4, 11) = 0$$
$$19k - 4 + 11 = 0$$
$$\boxed{k = \frac{-7}{19}}$$

$$\vec{n_1} \times \vec{n_2} = \begin{vmatrix} -2 & -1 \\ 9 & -5 \end{vmatrix} \mathbf{i} \begin{vmatrix} 1 & -1 \\ 1 & -5 \end{vmatrix} \mathbf{j} \begin{vmatrix} 1 & -2 \\ 1 & 9 \end{vmatrix} \mathbf{k}$$
$$= (19, 4, 11)$$